

Lecture 12

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We consider a (n, k) binary linear code and analyze the weight distribution. It is denoted by A_i for $i \in \{0, 1, \dots, n\}$, the number of codewords with weight i . The weight enumeration polynomial $A(z)$ is defined as:

$$A(z) = A_0 + A_1 z + A_2 z^2 + \dots + A_n z^n \quad (1)$$

The weight enumeration polynomial for the dual code is denoted by $B(z)$.

1.1 Relationship between $A(z)$ and $B(z)$

$$\begin{aligned} B(z) &= \frac{1}{2^k} \sum_{i=0}^n A_i \cdot (1-z)^i \cdot (1+z)^{n-i} \\ &= \frac{1}{2^k} (1+z)^n A\left(\frac{1-z}{1+z}\right) \end{aligned} \quad (2)$$

We can rewrite the above equations as:

$$B_j = \frac{1}{2^k} \sum_{i=0}^n A_i K_{ij}^n \text{ for } j \in \{0, 1, \dots, n\} \quad (3)$$

where

$$K_{ij}^n = \sum_{h=0}^i (-1)^h \cdot \binom{i}{h} \cdot \binom{n-i}{j-h} \quad (4)$$

Example: Parity check code $(3, 2)$.

The weight distribution is $A_0 = 1, A_1 = 0, A_2 = 3, A_3 = 0$, therefore $A(z) = 1 + 3z^2$.

Now, we construct the weight enumeration polynomial for the dual code $B(z)$.

$$\begin{aligned} B(z) &= \frac{1}{4} \cdot (1+z)^3 \cdot \left(1 + 3 \cdot \left(\frac{1-z}{1+z}\right)^2\right) \\ &= 1 + z^3 \end{aligned} \quad (5)$$

Therefore, the weight distribution for the dual code is $[1, 0, 0, 1]$, corresponding to the $(3, 1)$ repetition code. This confirms the fact that the dual code of parity check code is indeed the repetition code.

1.2 Proof of the identity

Before we begin the proof, we define a function $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \mathbb{R}$ as:

$$f(x, y) = (-1)^{\langle x|y \rangle}. \quad (6)$$

Here, $\langle \cdot | \cdot \rangle$ denotes the inner product defined by XOR of component-wise AND.

Lemma 1: For (n, k) linear code C , and $y \in \{0, 1\}^n$, we have:

$$\sum_{x \in C} f(x, y) = \begin{cases} 2^k & \text{if } y \in C^\perp \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

Proof: For the case when $y \in C^\perp$, we have $\langle x|y \rangle = 0, \forall x \in C$. Therefore, $f(x, y) = 1$. As $|C| = 2^k$, we have the summation to be 2^k .

For case when $y \notin C^\perp$, then there exists x_0 s.t. $f(x_0, y) = -1$. We also observe that $f(x + x_0, y) = -f(x, y)$, using the fact that the inner product is linear for the first argument.

We know that the set C is the same as the set $x_0 + C$. Hence we have

$$\begin{aligned} \sum_{x \in C} f(x, y) &= \sum_{x \in C} f(x + x_0, y) \\ &= - \sum_{x \in C} f(x, y) \end{aligned} \quad (8)$$

Therefore, we have $\sum_{x \in C} f(x, y) = 0$ for this case. Hence proved.

Lemma 2: Consider $x \in \{0, 1\}^n$ with $w(x) = i$ (Hamming weight), and V_j containing all vectors in $\{0, 1\}^n$ with Hamming weight j .

$$\begin{aligned} \sum_{y \in V_j} f(x, y) &= K_{ij}^{(n)} \\ &= \sum_{h=0}^j (-1)^h \cdot \binom{i}{h} \cdot \binom{n-i}{j-h} \end{aligned} \quad (9)$$

Proof: Without loss of generality, we assume

$$x = (\underbrace{11 \dots 1}_{i \text{ times}} \underbrace{00 \dots 0}_{n-i \text{ times}}). \quad (10)$$

The number of vectors that share h 1s with x will have h 1s in the first h positions and the remaining j 1s in the last $n - i$ positions, which is $\binom{i}{h} \cdot \binom{n-i}{j-h}$. And for each such y , we will have $f(x, y) = (-1)^h$. Therefore, we have

$$\sum_{y \in V_j} f(x, y) = \sum_{h=0}^j (-1)^h \cdot \binom{i}{h} \cdot \binom{n-i}{j-h} \quad (11)$$

Hence proved.

Now, coming to the proof of the original identity.

Proof: Using Lemma 1, we have:

$$\begin{aligned}
\sum_{y \in V_j} \sum_{x \in C} f(x, y) &= \sum_{y \in V_j \cap C^\perp} \sum_{x \in C} f(x, y) \\
&= \sum_{y \in V_j \cap C^\perp} 2^k \\
&= B_j \cdot 2^k
\end{aligned} \tag{12}$$

Using Lemma 2, we have:

$$\begin{aligned}
\sum_{x \in C} \sum_{y \in V_j} f(x, y) &= \sum_{x \in C} K_{w(x)j}^{(n)} \\
&= \sum_{i=0}^n A_i K_{ij}^{(n)}
\end{aligned} \tag{13}$$

Combining equations 12 and 13, we get:

$$B_j = \frac{1}{2^k} \sum_{i=0}^n A_i K_{ij}^n \text{ for } j \in \{0, 1, \dots, n\}. \tag{14}$$

QED

2 Applications of McWilliam's Identity

We will look at three applications of the identity:

- Finding the probability of undetected errors over a BSC(p)
- Finding the probability of decoding errors over a BSC(p)
- Getting bounds on code parameters

2.1 Probability of undetected errors over a BSC(p)

For any codeword the probability that an error would go undetected is the probability that the error pattern lies in the code. Hence, we have the probability that the error would go undetected ($\mathbb{P}_u$)

$$\begin{aligned}
\mathbb{P}_u &= \sum_{c \in C \setminus \mathbf{0}} p^{w(c)} (1-p)^{n-w(c)} \\
&= \sum_{i=0}^n A_i p^i (1-p)^{n-i} \\
&= (1-p)^n \left[A\left(\frac{p}{1-p}\right) - 1 \right]
\end{aligned}$$

We now use MacWilliam's identity with $z = 1 - 2p$ to get

$$\mathbb{P}_u = \frac{1}{2^{n-k}} B(1 - 2p) - (1 - p)^n$$

This is useful in code families where the dual has a nice structure. An example is the $(7, 4)$ Hamming code. Using its parity check matrix, we get $B(z) = 1 + 7z^4$. Thus, $\mathbb{P}_u = \frac{1}{2^3}(1 + 7(1 - 2p)^4) - (1 - p)^7$.

2.2 Probability of decoding errors over a BSC(p)

We demonstrate an upper bound on the probability of decoding an error over the BSC(p) channel assuming a maximum likelihood based decoder. This decoder is given by

$$\mathcal{D}(y) = \operatorname{argmax}_{c \in C} \mathbb{P}(y|c)$$

Without loss of generality, we consider the transmitted codeword as $c = \mathbf{0}$. Then,

$$\begin{aligned} \mathbb{P}_e &= \sum_{c \in C \setminus \mathbf{0}} \mathbb{P}(\hat{c} = c) \\ &= \sum_{c \in C \setminus \mathbf{0}} \sum_{y \in \mathcal{D}(c)} \mathbb{P}(y|\mathbf{0}) \\ &\leq \sum_{c \in C \setminus \mathbf{0}} \sum_{y \in \mathcal{D}(c)} \sqrt{\mathbb{P}(y|\mathbf{0})\mathbb{P}(y|c)} \quad [\text{Since } c = \mathcal{D}(y)] \\ &= \sum_{c \in C \setminus \mathbf{0}} \sum_{y \in \mathcal{D}(c)} \prod_{i=1}^n \sqrt{\mathbb{P}(y_i|0)\mathbb{P}(y_i|c_i)} \quad [\text{Since it is BSC}] \end{aligned}$$

Now, instead of just considering the $y \in \mathcal{D}(c)$, we upper bound this by extending it to all y , to get

$$\begin{aligned} \mathbb{P}_e &\leq \sum_{c \in C \setminus \mathbf{0}} \prod_{i=1}^n \sum_{w \in \{0,1\}} \sqrt{\mathbb{P}(w|0)\mathbb{P}(w|c_i)} \\ &= \sum_{c \in C \setminus \mathbf{0}} \prod_{j: c_j \neq 0} \sum_{w \in \{0,1\}} \sqrt{\mathbb{P}(w|0)\mathbb{P}(w|1)} \end{aligned}$$

We define the Bhattacharya number of the channel $\gamma = \sum_{w \in \{0,1\}} \sqrt{\mathbb{P}(w|0)\mathbb{P}(w|1)}$. For a BSC, this evaluates to $\gamma = 2\sqrt{p(1-p)}$. For a BEC, we have $\gamma = p$. Continuing with the BSC assumption, we get

$$\begin{aligned} \mathbb{P}_e &\leq \sum_{c \in C \setminus \mathbf{0}} \gamma^{w(c)} \\ &= A(\gamma) - 1 \end{aligned}$$

This finally gives us $\mathbb{P}_e \leq A(\gamma) - 1$.

2.3 Bounds on code parameters

We note in the MacWilliam's identity that $K_{0j}^{(n)} = \binom{n}{j}$. Also, since $B_j \geq 0$, we get $\sum_{i=0}^n A_i K_{ij}^{(n)} \geq 0$. Since, $A_i = 0 \ \forall i \in \{1, 2, \dots, d-1\}$, we have $\sum_{i=d}^n A_i K_{ij}^{(n)} \geq -\binom{n}{j}$.